SMART UP ONLINE — FEBRUARY 28 DELIVERY PLAN

Feature-Complete MVP — Client Submission

Document Type: Phase 1 MVP — Essential Features + Advanced Capabilities

Version: 1.0 (Expanded Scope)

Delivery Deadline: February 28, 2026 (8 Days Building Time)

Platform Name: SmartUp Education Platform

Target Capacity: 100 Students Across Multiple Rooms

Scope: Core classroom functionality + AI monitoring + Payment gateway + Exam system + Ghost Mode + WhatsApp notifications + HR/Payroll

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1. Executive Summary

This document defines the **Minimum Viable Product (MVP)** to be delivered by **February 28, 2026** for client submission.

What This Delivers

✅ **Live online classroom system** capable of running **100 concurrent students across**

multiple rooms

✅ **Three-server architecture**: Frappe ERP + SmartUp Next.js + LiveKit Media

✅ **6 essential user roles**: Owner, Batch Coordinator, Academic Operator, Teacher, Student, Parent

✅ **Core classroom features**: HD video/audio, screen share, chat, whiteboard

✅ **AI-powered student monitoring** using MediaPipe (attention, drowsiness, distraction detection)

✅ **Ghost Mode** for silent classroom observation by Owner AND Batch Coordinator

✅ **Complete HR/Payroll system** (salary structures, LOP calculation, incentives, performance tracking, payslip generation)

✅ **Federal Bank payment gateway** with period-wise billing enforcement

✅ **Basic recording** to MinIO storage with playback access

✅ **Complete exam & assessment system** (MCQ + descriptive, auto-grading, results)

✅ **WhatsApp notifications** for critical alerts (class reminders, payment due, attendance)

✅ **Basic scheduling and batch management**

✅ **Automatic attendance tracking**

✅ **Role-specific dashboards**

✅ **Secure token-based room access**

✅ **Recording to MinIO storage** with playback access

What This Does NOT Include (Phase 2-4)

❌ YouTube live streaming with video access request system (Phase 2)

❌ Full CRM with lead management (basic lead tracking only in Phase 1)

❌ Complex reporting (14 reports - only 5 essential reports in Phase 1, rest in Phase 2-3)

❌ Multi-language/Arabic support (English only for Phase 1, Arabic in Phase 3)

❌ Advanced analytics dashboards (Phase 3)

❌ Mobile apps for Student/Parent (Phase 4)

❌ Advanced homework submission with grading workflow (Phase 2)

❌ Content management system with syllabus library (Phase 3)

❌ Predictive analytics using ML (Phase 4)

Success Criteria

The system must successfully:

1. Support **5 concurrent classrooms** with **20 students each** (100 total)
2. Allow teachers to **start classes** from Frappe
3. Allow students to **join classes** only after payment (period-wise billing enforcement)
4. Deliver **HD video/audio** with acceptable quality (<2% connection drops)
5. **AI monitor students** in real-time (drowsiness, distraction, absence detection with >90% accuracy)
6. **Track attendance automatically** (join/leave times with 100% accuracy)
7. **Process payments** via Federal Bank gateway (>95% success rate, multi-currency GCC)
8. Allow **Ghost Mode observation** by Owner AND Batch Coordinator (100% invisible to participants)
9. Deliver **MCQ exams** with auto-grading (100% accuracy) and instant results
10. **Record classes to MinIO** storage with 100% success rate and playback access
11. Send **WhatsApp notifications** for critical alerts (>95% delivery rate)
12. **Process HR/Payroll** with LOP calculation, incentives, and auto-generated payslips
13. Display **role-specific dashboards** for all 6 roles with real-time data
14. Run **stable for 60-minute classes** without crashes or performance degradation

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3. Multi-Phase Roadmap (4 Phases)

Phase 1: February 28, 2026 (8 Days Building Time)

Focus: Core platform with live classroom, payments, exams, AI monitoring, HR/Payroll

What's Included: ✅ 6 user roles with dashboards

✅ Live classroom with AI monitoring + Ghost Mode (Owner + Coordinator)

✅ Federal Bank payment gateway with period-wise billing

✅ Recording to MinIO storage with playback access

✅ Complete exam system (MCQ auto-grading)

✅ Complete HR/Payroll system (salary, LOP, incentives, payslips)

✅ WhatsApp notifications

✅ 5 essential reports (Daily Business, Weekly Sales, Monthly Academic, Monthly Revenue, Teacher Performance)

✅ Basic batch management, scheduling, attendance

✅ 100 concurrent students capacity

Phase 2: Post Phase 1

Focus: YouTube streaming + Video access + CRM + Advanced operational features

What Will Be Built:

#	Feature Area	Details
1	**YouTube Live Streaming**	RTMP integration with YouTube Data API v3, automatic stream creation, real-time stream health monitoring, unlimited cloud storage
2	**Video Access Request System**	Student requests access to recorded videos, batch coordinator/teacher approval workflow, access logs and analytics
3	**Full CRM System**	Complete lead funnel, call monitoring, call recording storage, demo management, sales executive performance tracking, Meta/Google Ads integration
4	**Homework Submission**	File uploads (PDF/images),

	System	teacher grading interface, deadline tracking, late submission penalties, homework analytics
5	**Advanced Reporting (5 more)**	Student Progress Report, Region-wise Growth, Drop-out & Retention, Complaint Resolution, Batch Profitability
6	**Doubt Clearing System**	Subject-wise doubt submission, file attachments, doubt resolution tracking, live doubt sessions
7	**Leave Management**	Student leave requests, approval workflow, leave balance, parent notifications
8	**Complaint Escalation**	Multi-level escalation (Coordinator → Operator → Owner), SLA tracking, resolution analytics
9	**Parent-Teacher Meeting**	Online PTM booking, time slot selection, meeting history, feedback collection
10	**Enhanced Student Dashboard**	Gamification (badges, leaderboard), achievement tracking, performance alerts

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Phase 3: Post Phase 2

Focus: Advanced features + Multi-language + Analytics + Content management

What Will Be Built:

#	Feature Area	Details
1	**Multi-Language Support**	Arabic/RTL support, language switcher, translation system, GCC localization (date formats, currencies)
2	**Advanced Digital Whiteboard**	Shapes, layers, collaborative drawing, save/load boards, text annotations, image insertion
3	**Content Management System**	Syllabus library, lesson planning tools, curriculum tracking, teaching material repository
4	**Advanced Analytics Dashboards**	Predictive analytics, student performance prediction, drop-out risk analysis, revenue forecasting
5	**Remaining Reports (4 more)**	HR & Working Hours, Communication Report, System Usage, Refund Analytics
6	**Descriptive Exam Grading**	Upload answer sheets, manual grading interface, partial marking, subjective question bank
7	**Advanced Parent Portal**	Multi-child management, complaint escalation, payment history analytics,

		performance comparison
8	**Social Media Integration**	Media team dashboard, content calendar, campaign tracking, creative performance analytics

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Phase 4: Post Phase 3

Focus: Mobile apps + Advanced features + Scaling

What Will Be Built:

#	Feature Area	Details
1	**Mobile Apps (Student)**	Native iOS/Android app. Features: Join class (seamless video), Make payments (in-app gateway), View reports/results, Take MCQ exams, Submit homework, View attendance, Receive push notifications, Offline content access
2	**Mobile Apps (Parent)**	Native iOS/Android app. Features: View child dashboard, Make payments, View reports/results, Attendance tracking, Receive push notifications, Multi-child support, Fee payment history
3	**Advanced Attendance Tracking**	Partial attendance, late arrival tracking, early exit reasons, attendance regularization, biometric

		integration (future-ready)
4	**Third-Party Integrations**	Google Classroom sync, Microsoft Teams integration, Zoom backup option, Calendar sync (Google/Outlook)
5	**ML-Based Features**	Student performance prediction, drop-out risk AI model, personalized learning recommendations, automatic weak area identification
6	**Advanced Security**	Two-factor authentication (2FA), IP whitelisting, session management, device tracking, suspicious activity alerts
7	**Scalability Enhancements**	Auto-scaling infrastructure, CDN integration, database optimization, caching layers, load balancing
8	**API Marketplace**	Public APIs for third-party integrations, developer documentation, webhook support, rate limiting

Mobile App Features (Detailed):

Student Mobile App:

- ✅ One-tap class join (seamless LiveKit mobile SDK)
- ✅ Federal Bank payment gateway (mobile-optimized)
- ✅ View exam results and report cards
- ✅ Take MCQ exams on mobile
- ✅ Submit homework (camera + file upload)
- ✅ View attendance percentage

- ✅ Push notifications (class reminders, results)
- ✅ Offline mode (download study materials)
- ✅ Dark mode support

Parent Mobile App:

- ✅ View child's schedule and attendance
- ✅ Make fee payments (mobile-optimized gateway)
- ✅ View exam results and reports
- ✅ Multi-child dashboard (switch between children)
- ✅ Push notifications (absence, low grades, payment due)
- ✅ Communication with Batch Coordinator
- ✅ Payment history and invoices
- ✅ Dark mode support

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2. Scope — What We Include (Phase 1: Feb 28 Delivery)

#	Feature Area	What's Included
1	**User Management**	Login system, 6 roles (Owner/Coordinator/Operator/Teacher/Student/Parent), role-based permissions
2	**Batch Management**	Create batches, assign teacher/coordinator, add students, set capacity, approval workflow
3	**Scheduling**	Class scheduler, calendar view, batch assignment, conflict detection
4	**Live Classroom**	HD video/audio, screen share, chat, whiteboard, hand-raise, AI monitoring
5	**AI Monitoring**	MediaPipe face detection, attention tracking, drowsiness alerts, real-time

		teacher alerts
6	**Ghost Mode**	Silent observation by Owner + Batch Coordinator, invisible entry, full classroom visibility, audit logging
7	**Attendance**	Auto-capture join/leave times, duration tracking, reports, parent alerts
8	**Recording**	MinIO storage with playback access, metadata tagging, basic access control
9	**Payment Gateway**	Federal Bank integration, multi-currency GCC, period-wise billing, class access enforcement
10	**Exam System**	MCQ + descriptive exams, question bank, auto-grading, results, report cards
11	**HR/Payroll System**	Salary structures, LOP calculation, teacher payment (per-hour/fixed), incentives, payslip generation
12	**Dashboards**	Role-specific dashboards with real-time widgets (6 roles)
13	**Notifications**	In-app + Email + WhatsApp (class reminders, payments,

		attendance, emergencies)
14	**Multi-Room**	Support 5-10 concurrent rooms
15	**Capacity**	100 concurrent students total across all rooms

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4. Three-Server Architecture (Simplified)

Server Specifications (Minimum)

Server	CPU	RAM	Storage	Purpose
LiveKit Media	8 vCPU	16 GB	500 GB SSD	Video/audio processing, recording storage
SmartUp Next.js	4 vCPU	8 GB	50 GB SSD	Classroom UI, API, token generation

Note: Frappe ERP is already set up and running. Only LiveKit and SmartUp servers need fresh configuration.

Network: Each server needs minimum 1 Gbps connection. LiveKit server needs low latency (<100ms to GCC region).

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5. Essential Roles (6 Core Roles)

#	Role	Login URL	Primary Functions
1	**Owner/Admin**	app.smartup.live	Full system access, user management, batch approval, ghost mode, monitor all classes
2	**Batch Coordinator**	app.smartup.live	Manage assigned batches, approve video access requests, monitor attendance, communicate with parents
3	**Academic Operator**	app.smartup.live	Day-to-day operations, schedule management, attendance monitoring, homework tracking
4	**Teacher**	app.smartup.live	Start classes, view schedule, check attendance, grade exams, access recordings
5	**Student**	app.smartup.live	Join classes (after payment), view schedule, take exams, request video access, submit homework
6	**Parent**	app.smartup.live	View child's

			dashboard, make payments, view reports, receive alerts

Deferred Roles (Phase 2):

- Sales & Marketing GM
- Accounts Department (full financial module)
- HR Associate (full HR module)
- Media Team
- Sales Executive
- Academic Coordinator (strategic planning - Operator handles tactical)

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6. Core Features — Frappe ERP

6.1 User Management

Feature	Description
User Creation	Admin creates users: Owner, Teacher, Student, Parent
Role Assignment	Assign one of 4 roles per user
Login System	Email + password authentication (Frappe built-in)
Password Reset	Email-based password reset
Session Management	Secure session handling

6.2 Batch Management

Feature	Description
Create Batch	Define: Batch Name, Subject, Teacher, Max

	Students, Category (1-to-1 / 1-to-5 / Group)
Assign Students	Add students to batch (individual or bulk upload via Excel)
Batch Status	Active / Inactive toggle
Capacity Alert	Warning when batch reaches max capacity
Teacher Assignment	Link one teacher per batch

Frappe DocType: Batch

- Fields: name, subject, teacher (link), max_students, current_students, status, category

6.3 Class Scheduling

Feature	Description
Create Schedule	Define: Batch, Date, Start Time, Duration (minutes), Room Name
Calendar View	Visual calendar showing all scheduled classes
Recurring Schedules	Define repeating pattern (e.g., Sun/Tue/Thu at 5 PM)
Conflict Detection	Prevent double-booking same teacher at same time
Status Tracking	Scheduled / Live / Completed / Cancelled

Frappe DocType: Class Schedule

- Fields: batch (link), date, start_time, duration, teacher (link), status, room_url

6.4 Attendance Tracking

Feature	Description
Auto-Capture	SmartUp pushes attendance data via API after class
Fields Captured	Student, Class, Join Time, Leave Time, Duration, Status (Present/Absent/Early Exit)
Reports	Daily attendance report, Student-wise attendance %, Batch-wise attendance
Manual Override	Admin can correct attendance records

Frappe DocType: Attendance

- Fields: student (link), class_schedule (link), join_time, leave_time, duration, status

6.5 Recording Management

Feature	Description
Auto-Record	LiveKit auto-records every class
Storage	MinIO on Server 2
Metadata	Recording linked to Class Schedule with file path, size, duration
Playback	Direct playback link available in Frappe
Access Control	Teacher can access all batch recordings, Students can access own class recordings

Frappe DocType: Class Recording

- Fields: class_schedule (link), file_path, file_size, duration, recorded_at

6.6 Dashboards

Owner/Admin Dashboard

- **Total Students**: Count of active students
- **Total Batches**: Count of active batches
- **Today's Classes**: List of scheduled classes with status
- **Currently Live**: Classes happening now with "Enter Ghost Mode" button
- **Revenue Today**: Fees collected today
- **Payment Pending**: Count of students with overdue payments
- **AI Alerts**: Summary of attention issues flagged today
- **Quick Actions**: Create Batch, Create Schedule, View All Attendance, Enter Ghost Mode

Batch Coordinator Dashboard

- **My Batches**: Assigned batches with student count and status
- **Today's Classes**: Classes for my batches with "Monitor" and "Enter Ghost Mode" buttons
- **Pending Video Requests**: Student video access requests awaiting approval
- **Low Attendance Alerts**: Students below attendance threshold
- **Recent Feedback**: Latest class feedback from students
- **Parent Messages**: Recent communication from parents
- **Quick Actions**: Approve Video Access, Enter Ghost Mode, Send Announcement

Academic Operator Dashboard

- **Today's Schedule**: All classes with status (upcoming/live/completed)
- **Teacher Readiness**: Teachers online/offline status
- **Attendance Summary**: Real-time attendance across all batches
- **Pending Tasks**: Rescheduling requests, leave applications
- **Exam Schedule**: Upcoming exams needing attention
- **System Alerts**: Technical or operational issues
- **Quick Actions**: Reschedule Class, Process Leave, Send Notifications

Teacher Dashboard

- **Today's Classes**: My scheduled classes with "Start Class" button
- **This Week**: Calendar view of upcoming classes
- **My Batches**: List of assigned batches with student count
- **Recent Attendance**: Latest attendance records for my classes
- **AI Monitoring**: Attention summary from recent classes
- **Pending Tasks**: Exams to grade, homework to review
- **Quick Actions**: Start Class, View Schedule, Grade Exams, View Recordings

Student Dashboard

- **Today's Classes**: My scheduled classes with "Join Class" button (green if paid, red if payment due)
- **Payment Status**: Current period status with "Pay Now" button if pending

- **This Week**: Calendar view of upcoming classes
- **My Attendance**: Current attendance percentage
- **Upcoming Exams**: Next exams with countdown
- **Recent Results**: Latest exam scores
- **Recording Library**: Available recordings and pending access requests
- **Quick Actions**: Join Class, Make Payment, Request Video Access, Take Exam

Parent Dashboard

- **Child's Schedule**: Today's and this week's classes
- **Payment Status**: Fee status with "Pay Now" button
- **Attendance %**: Child's current attendance percentage
- **Recent Results**: Latest exam scores and rank
- **AI Monitoring**: Attention summary (if enabled)
- **Recent Classes**: Last 5 classes with recordings
- **Alerts**: Important notifications (absence, low grades, payment due)
- **Quick Actions**: Make Payment, View Full Report, View Recordings

6.7 Payment Gateway Integration

Federal Bank Gateway Configuration

Setting	Value
Provider	Federal Bank Payment Gateway
Integration	Frappe custom Payment Gateway DocType
Currencies	AED, QAR, SAR, OMR, KWD, BHD
Methods	Credit/Debit Card, Digital Wallets
Mode	Redirect to Federal Bank hosted page

Period-Wise Billing Enforcement

Payment Flow:

Student clicks "Join Class" in Frappe

```
│
├─ Frappe checks: Is current period payment done?
│
├─ IF PAID:
│   ├─ Generate SmartUp token
│   └─ Open classroom (new tab)
│
└─ IF NOT PAID:
    ├─ Show Payment Screen
    ├─ Redirect to Federal Bank Gateway
    ├─ Complete payment
    ├─ Federal Bank callback → Frappe
    ├─ Frappe marks as PAID
    └─ Now "Join Class" button active
```

Frappe DocTypes:

- **Fee Structure**: Batch, Amount, Currency, Period (Monthly/Quarterly)
- **Payment Entry**: Student, Amount, Period, Status, Gateway Response
- **Payment Period**: Student, Start Date, End Date, Status (Paid/Pending/Overdue)

6.8 Exam & Assessment System

MCQ Exam Features

Feature	Description
Question Bank	Subject-wise repository with difficulty tagging
Exam Creation	Teacher creates: Subject, Duration, Total Marks, Questions (random or manual)
Timer	Countdown timer with auto-submit
Anti-Cheating	Question randomization per student
Auto-Grading	Instant scoring for MCQ
Results	Marks, percentage, rank, analysis

Frappe DocTypes:

- **Question Bank**: Subject, Chapter, Question, Options (A/B/C/D), Correct Answer, Difficulty
- **Exam**: Batch, Subject, Date, Duration, Total Marks, Status
- **Exam Submission**: Student, Exam, Answers (JSON), Score, Rank, Submitted At
- **Result**: Student, Exam, Marks, Percentage, Rank, Report Card (PDF)

Descriptive Exam

- Teacher uploads question paper (PDF)
- Student uploads answer sheet (PDF/images)
- Teacher manually grades and enters marks

6.9 HR/Payroll System (Complete)

Salary Structure Setup

Feature	Description
Salary Types	Fixed Monthly, Per-Hour Rate, Per-Class Rate, Hybrid (Fixed + Per-Class)
Payment Period	Monthly, Bi-monthly, Custom period
Auto-Calculation	System calculates based on actual classes conducted (from attendance logs)
LOP Deduction	Automatic Loss of Pay based on teacher attendance/absence
Incentive Rules	Performance-based incentives (attendance %, student feedback score, batch results)

Frappe DocTypes:

- **Salary Structure**: Employee, Salary Type, Base Amount, Per-Hour Rate, Per-Class Rate, Payment Period
- **Salary Component**: Basic, HRA, Incentive, LOP Deduction, Net Payable
- **Payroll Entry**: Employee, Period, Classes Conducted, Working Days, LOP Days, Incentive Amount, Total Salary
- **Payslip**: Auto-generated PDF with breakdown, downloadable by teacher

Payroll Processing Workflow

```
End of Payment Period (e.g., Month End)
  │
  ├─ System fetches teacher attendance data (from SmartUp API)
  ├─ System counts classes conducted per teacher
  ├─ System calculates working hours (if per-hour)
  ├─ System applies LOP deduction (absences, late logins)
  ├─ System calculates incentives:
  │   ├─ Attendance bonus (if >95% attendance)
  │   ├─ Feedback bonus (if avg rating >4.5)
  │   └─ Result bonus (if batch average >80%)
  │
  ├─ Accounts Department reviews payroll
  ├─ Admin approves payroll
  │
  ├─ Payslips auto-generated
  ├─ Email sent to teachers with payslip PDF
  └─ Payment processed (manual bank transfer or integrated)
```

Teacher Performance Tracking

Metric	Calculation	Impact
Attendance %	Classes conducted / Classes scheduled × 100	LOP if <95%
Average Class Rating	Sum of student ratings / Total ratings	Bonus if >4.5
Batch Performance	Batch average exam score	Bonus if >80%
Punctuality	On-time class starts / Total classes × 100	Warning if <90%
Student Retention	Students remaining / Initial students × 100	Bonus if >95%

Monthly Report: Teacher Performance Report (auto-generated, used for appraisals)

6.10 Notifications (Multi-Channel)

Trigger	Recipients	Channels
Class Reminder (30 min)	Teacher, Student, Parent	In-App + Email + WhatsApp
Class Reminder (5 min)	Teacher, Student	In-App
Payment Due (7 days)	Student, Parent	In-App + Email + WhatsApp
Payment Due (1 day)	Student, Parent	In-App + Email + WhatsApp
Payment Overdue	Student, Parent	In-App + Email + WhatsApp
Payment Confirmed	Student, Parent	In-App + Email
Student Absent	Parent	In-App + Email + WhatsApp
Low Attendance Alert	Parent, Coordinator	In-App + Email + WhatsApp
AI Alert (Drowsy/Distracted)	Teacher (real-time)	In-Class UI
Exam Scheduled	Student, Parent	In-App + Email
Results Published	Student, Parent	In-App + Email + WhatsApp
Video Access Approved	Student	In-App + Email
New Schedule	Teacher, Students	In-App + Email

WhatsApp Integration:

- Frappe WhatsApp integration via Twilio or direct WhatsApp Business API
- Template messages for compliance
- Only critical notifications via WhatsApp to avoid spam

7. Core Features — SmartUp Classroom

7.1 Pre-Join Lobby

Feature	Description
Token Validation	Validate JWT token from URL, extract user identity and room permission
Device Selection	Select camera and microphone
Preview	Test video/audio before joining
Join Button	Connect to LiveKit room

7.2 Live Classroom — Teacher View

Feature	Description
HD Video	Teacher's camera broadcast to all students (720p minimum, 1080p ideal)
HD Audio	Clear audio with echo cancellation
Screen Share	Share entire screen or application window
Digital Whiteboard	Basic drawing canvas with pen, colors, eraser (synced in real-time)
In-Class Chat	Real-time text chat with all students
Participant List	See all connected students with names, connection quality, AI status badge
Hand-Raise Queue	Ordered list of students requesting to speak. Click to grant permission

Mute Controls	Mute/unmute individual students
AI Alert Panel	Real-time alerts: Which students are drowsy/distracted/absent. Color-coded badges (Green/Yellow/Red)
Class Timer	Visual countdown showing time remaining in scheduled class
Recording Indicator	Red dot when recording + "LIVE" badge when streaming to YouTube
YouTube Status	YouTube stream health indicator
Connection Quality	Visual network quality indicator
Controls Bar	Mic toggle, Camera toggle, Screen Share, Whiteboard, Chat, Participants, Hand-Raise Queue, End Class
End Class	Ends session, stops YouTube stream, triggers post-class data sync to Frappe

7.3 Live Classroom — Student View

Feature	Description
Teacher Video	Full-screen view of teacher's camera
Screen View	See teacher's screen share
Whiteboard View	See teacher's whiteboard annotations in real-time
Chat	Send messages, ask questions
Hand Raise	Click button to raise hand, enter queue

Mic/Camera	Enable if teacher grants permission (default: off)
AI Attention Indicator	Discreet personal indicator showing own attention score. Gentle prompts if distracted ("Please focus on screen")
Leave Class	Exit classroom
Connection Indicator	Visual network quality indicator

7.4 Live Classroom — Ghost Mode View (Owner + Batch Coordinator)

Feature	Description
Invisible Entry	Owner OR Batch Coordinator can enter any live classroom without appearing in participant list
Full Visibility	Can see/hear everything: teacher video, student videos, screen share, whiteboard, chat
AI Panel Access	View all student attention data and alerts
Participant List	See all participants — but ghost is NOT visible to them
No Interaction	Cannot send video, audio, chat, or interact in any way
Ghost Indicator	Subtle overlay (visible only to observer) confirming ghost mode active
Exit Button	Only control available — exit and return to Frappe
Audit Logged	Entry time, duration, exit time logged in

	Frappe for compliance
Access Control	Owner can access ALL classrooms. Batch Coordinator can ONLY access their assigned batch classrooms

How It Works:

```
Owner/Batch Coordinator in Frappe → Views "Currently Live" classes widget
  │
  ├─ Owner: Sees ALL live classes with "Enter Ghost Mode" button
  ├─ Batch Coordinator: Sees ONLY assigned batch classes with "Enter Ghost Mode" button
  │
  ├─ Clicks "Enter Ghost Mode" next to a live class
  │
  ├─ Frappe calls SmartUp API: POST /api/generate-ghost-token
  │   └─ Payload: {user_id, user_role, room_name, role: "ghost"}
  │
  ├─ SmartUp validates:
  │   ├─ If Owner → Allow access to any room
  │   └─ If Batch Coordinator → Verify room belongs to assigned batch
  │
  ├─ SmartUp generates special JWT token with hidden=true
  │
  ├─ New tab opens: class.smartup.live/ghost?token=xxx
  │
  ├─ LiveKit connects observer with invisible permissions
  │
  └─ Observer monitors silently, nobody knows they're there
```

Use Cases:

- **Owner**: System-wide quality monitoring, teacher performance evaluation, compliance checks
- **Batch Coordinator**: Monitor assigned batch classes, verify teaching quality, observe student engagement, gather feedback data

7.5 AI-Powered Student Monitoring (MediaPipe)

How It Works

Google MediaPipe Face Mesh runs **client-side in student's browser** (WebAssembly). Zero video transmission.

Layer	What Happens
Student Browser	Camera feed analyzed by MediaPipe at 5 FPS
Face Mesh	468 facial landmark points detected
Eye Analysis	Eye Aspect Ratio (EAR) — detects closed eyes (drowsiness)
Head Pose	Yaw/Pitch angles — detects looking away (distraction)
Presence	Face detection — detects if student visible (absence)
Alert Generation	JSON alert sent via LiveKit data channel when thresholds exceeded
Teacher View	SmartUp displays real-time alerts on teacher's panel
Data Sync	Alert summary pushed to Frappe after class for reports

Detection Thresholds

Metric	Method	Threshold	Alert Trigger
Drowsiness	Eye Aspect Ratio (EAR)	EAR < 0.21	Eyes closed > 3 seconds
Distraction	Head yaw/pitch	Yaw > ±30° or Pitch > ±25°	Looking away > 4 seconds

Absence	Face not detected	No face found	Not visible > 5 seconds

What Each Role Sees:

- **Student**: Personal attention indicator + gentle prompt ("Stay focused!")
- **Teacher**: Real-time badge per student (Green/Yellow/Red), alert count, class average
- **Admin**: Full AI monitoring report in Frappe (batch-wise, teacher-wise analytics)

Privacy:

- 100% client-side processing
- Zero video transmitted
- Only JSON alerts sent (type + duration + timestamp)
- Student consent on first use
- Data anonymized after 30 days

7.6 YouTube Live Streaming Integration

Dual Recording Strategy

Storage	Purpose	Access
YouTube	Primary — unlimited free storage, global CDN	Approval-based time-limited private links
MinIO (Server 2)	Backup — local control, faster access	Teacher/Admin direct access

How It Works

During Class:

1. LiveKit streams to YouTube via RTMP (rtmps://a.rtmp.youtube.com/live2)
2. Video codec: H.264 High Profile, 1080p @ 4500kbps or 720p @ 2500kbps, 30fps
3. Audio codec: AAC-LC, 128kbps stereo, 44.1kHz
4. YouTube processes stream and saves as private video
5. LiveKit simultaneously saves to MinIO as backup

After Class:

1. SmartUp waits 5-15 minutes for YouTube processing
2. SmartUp calls YouTube API to get video_id, watch_url, thumbnail
3. SmartUp auto-organizes into YouTube playlist (Batch + Subject + Month)

4. Metadata synced to Frappe: Subject, Batch, Date, Teacher, Duration, YouTube link, MinIO link

Student Video Access Request Workflow:

```
Student in Frappe → Recording Library
  │
  ├─ Browses available recordings (by subject/date)
  │
  ├─ Clicks "Request Access" → Enters reason
  │
  ├─ Request creates Frappe DocType: Video Access Request
  │   Fields: student, recording, reason, status (Pending), requested_at
  │
  ├─ Batch Coordinator sees request in dashboard
  │
  ├─ Coordinator reviews: student attendance, reason, recording details
  │
  ├─ Coordinator Approves:
  │   ├─ SmartUp generates time-limited YouTube private link (expires in 7 days)
  │   ├─ Link sent to student via notification
  │   └─ Access logged with expiry date
  │
  └─ OR Coordinator Rejects:
      └─ Reason entered, student notified
```

Frappe DocTypes:

- **Class Recording**: Enhanced with youtube_video_id, youtube_watch_url, youtube_playlist_id, minio_file_path
- **Video Access Request**: student, recording, request_reason, status, reviewed_by, approved_at, youtube_access_link, expires_at
- **YouTube Configuration**: channel_id, api_key, rtmp_url, default_privacy (private)

7.7 Room Types & Capacity

Room Type	Max Students	Student Visibility	Student Interaction

One-to-One	1	See teacher	Full mic/camera
One-to-Five	5	See teacher only	Hand-raise to speak
Group Batch	20	See teacher only	Hand-raise to speak
Lecture	50	See teacher only	Chat + hand-raise

Total Concurrent: 5 rooms × 20 students = **100 students** (or 10 rooms × 10 students, configurable)

7.8 Post-Class Data Sync

When class ends, SmartUp automatically pushes to Frappe via API:

Data	Format
Attendance	JSON array: [{student_id, join_time, leave_time, duration}, ...]
Recording Metadata	JSON: {minio_file_path, youtube_video_id, youtube_watch_url, file_size, duration, recorded_at}
AI Monitoring Summary	JSON per student: {student_id, attentive_%, distracted_%, drowsy_%, absent_%, alert_count}
Chat Log	JSON array of messages (stored in SmartUp, summary to Frappe)
YouTube Playlist	YouTube playlist_id for batch/subject organization

---

8. Core Features — LiveKit Media Server

8.1 What LiveKit Handles

Feature	Description
WebRTC Streaming	Real-time video/audio encoding and distribution
Simulcast	Multiple quality streams per participant (auto-adjusted by bandwidth)
Recording	Auto-record to MinIO object storage
Room Management	Create/destroy rooms via API
Token Validation	Verify JWT tokens for room access
Network Adaptation	Automatic bitrate adjustment based on connection quality

8.2 Recording Settings

Setting	Value
Codec	H.264 High Profile (video), Opus (audio for MinIO), AAC-LC (audio for YouTube)
Resolution	1280x720 (teacher), 640x480 (students if recorded)
Format	MP4 (MinIO), RTMP stream (YouTube)
Storage	MinIO (S3-compatible) + YouTube (private videos)
Retention	90 days (MinIO backup), unlimited (YouTube)

YouTube RTMP	rtmps://a.rtmp.youtube.com/live2
YouTube Bitrate	Video: 4500kbps (1080p) or 2500kbps (720p), Audio: 128kbps stereo

8.3 Quality Settings

Metric	Target
Video Bitrate	1.5 Mbps (teacher), 0.5 Mbps (student)
Audio Bitrate	64 kbps (mono) or 128 kbps (stereo)
Frame Rate	30 fps
Latency	<300ms peer-to-peer

---

9. Join Class Flow

Step-by-Step Process

```
1. Student logs into Frappe (app.smartup.live)
   │
   ├──→ Frappe dashboard loads
   │
2. Student sees "Today's Classes" widget
   │
   ├──→ Scheduled class shows "Join Class" button (green, active)
   │
3. Student clicks "Join Class"
   │
   ├──→ Frappe backend:
   │    ├─ Checks if class is live (status = "Live")
   │    ├─ If NOT live → Shows "Class not started yet"
   │    ├─ If live → Calls SmartUp API: POST /api/generate-token
   │    │    └─ Payload: {user_id, room_name, role: "participant"}
```

```
  │   │
  │   └─ SmartUp responds: {token_url: "class.smartup.live/join?token=eyJhb..."}
  │
4. Frappe opens new browser tab with token URL
  │
  └──→ class.smartup.live/join?token=eyJhb...
      │
      ├──→ SmartUp Pre-Join Lobby loads:
      │    ├─ Validates token (JWT signature, expiry, room permission)
      │    ├─ Shows device selection (camera, mic)
      │    ├─ Shows preview
      │    └─ Shows "Enter Classroom" button
      │
      ├──→ Student clicks "Enter Classroom"
      │
      └──→ SmartUp connects to LiveKit (media.smartup.live)
          │
          ├─ LiveKit validates token
          ├─ LiveKit adds student to room
          ├─ Video/audio streams start
          └─ Student sees teacher video, chat, controls
```

---

10. 8-Day Delivery Timeline

Day 1: Infrastructure Setup

Task	Owner	Deliverable
Provision 2 cloud servers	DevOps	2 running Ubuntu servers with SSH
Install LiveKit on Media Server	Backend	LiveKit running, API accessible
Install PostgreSQL + Redis on SmartUp Server	Backend	Databases ready for SmartUp
Install MinIO on Media Server	Backend	Object storage ready

Configure SSL certificates (2 servers)	DevOps	HTTPS enabled
Verify Frappe ERP connectivity	Backend	Frappe accessible, API endpoints verified

End of Day 1: All infrastructure live and accessible.

Day 2: Frappe Core DocTypes

Task	Owner	Deliverable
Create custom Frappe app "smartup_core"	Backend	App installed
Create DocTypes: Batch, Class Schedule, Attendance	Backend	Core academic DocTypes
Create DocTypes: Class Recording	Backend	Recording management
Create DocTypes: Question Bank, Exam, Exam Submission, Result	Backend	Exam system DocTypes
Create DocTypes: Fee Structure, Payment Entry, Payment Period	Backend	Payment DocTypes
Create DocTypes: Salary Structure, Payroll Entry, Payslip	Backend	HR/Payroll DocTypes
Setup 6 user roles with permissions	Backend	Roles configured

End of Day 2: All Frappe DocTypes created, roles configured.

Day 3: Frappe Dashboards & Payment Gateway

Task	Owner	Deliverable
Build 6 role-specific dashboards	Backend	All dashboards with widgets
Integrate Federal Bank Payment Gateway	Backend	Payment flow working (test mode)
Build period-wise billing logic	Backend	Class access enforcement based on payment
Build HR/Payroll processing logic	Backend	LOP calculation, incentive rules configured
Create notification system (in-app + email)	Backend	Basic notifications working

End of Day 3: Dashboards live, payment gateway integrated.

Day 4: SmartUp Next.js Setup & Pre-Join

Task	Owner	Deliverable
Setup Next.js 14 project on SmartUp Server	Frontend	Next.js running on class.smartup.live
Build Pre-Join Lobby page	Frontend	Token validation, device selection, preview
Integrate LiveKit React SDK	Frontend	Video/audio connection working
Build room connection logic	Frontend	Join room via token

Test basic room entry	QA	Can enter empty room

End of Day 4: Pre-join lobby working, LiveKit connection established.

Day 5: SmartUp Classroom UI

Task	Owner	Deliverable
Build Teacher View UI	Frontend	Video grid, controls, participant list
Build Student View UI	Frontend	Teacher video, limited controls
Implement Screen Share	Frontend	Teacher can share screen
Implement Chat component	Frontend	Real-time text chat
Implement Hand-Raise queue	Frontend	Queue visible to teacher
Implement Digital Whiteboard (basic)	Frontend	Basic annotation tools working

End of Day 5: Basic classroom UI working for teacher and student.

Day 6: AI Monitoring & Ghost Mode

Task	Owner	Deliverable
Integrate MediaPipe Face Mesh (WASM)	Frontend	Face detection running client-side
Build AI alert detection logic	Frontend	EAR, head pose, presence detection

Build teacher AI alert panel	Frontend	Real-time alerts displayed to teacher
Implement Ghost Mode token generation	Backend	Special hidden tokens (Owner + Coordinator access)
Build Ghost Mode UI	Frontend	Invisible observer view
Implement batch-level access control	Backend	Coordinator limited to assigned batches

End of Day 6: AI monitoring working, Ghost Mode functional.

Day 7: Recording & Exam System

Task	Owner	Deliverable
Configure LiveKit recording to MinIO	Backend	Auto-recording enabled, files saved to MinIO
Implement recording playback URLs	Backend	Signed URLs for secure playback
Build MCQ exam creation UI	Frontend	Teacher can create exams
Build student exam-taking UI	Frontend	Student can take MCQ exam with timer
Implement auto-grading logic	Backend	Instant results for MCQ
Build result display & report cards	Frontend	Results visible, PDF generation

End of Day 7: Recording working, exam system functional.

Day 8: WhatsApp & Final Integration

Task	Owner	Deliverable
Integrate WhatsApp (Twilio)	Backend	Critical notifications via WhatsApp
Setup notification templates	Backend	All trigger templates configured
Build 5 essential reports	Backend	Daily Business, Weekly Sales, Monthly Academic, Monthly Revenue, Teacher Performance
Implement HR/Payroll UI	Frontend	Payroll processing, payslip generation, teacher dashboard
End-to-end API integration testing	QA	All Frappe-SmartUp-LiveKit flows working

End of Day 8: All features integrated and functional.

Note: Testing and bug fixing happen continuously throughout Days 1-8, not as a separate phase. Each feature is tested immediately after implementation.

---

11. Final Delivery Package

Testing & Quality Assurance (Throughout 8 Days)

Test Type	Coverage
End-to-end flow testing	Complete user journey for all 6 roles
Multi-room load testing	5 rooms × 20 students = 100 concurrent

Payment gateway testing	Payment → class access enforcement
AI monitoring stress test	20 students simultaneously
Recording reliability test	60-minute classes without failures
Ghost Mode testing	Silent observation verified (Owner + Coordinator)
HR/Payroll calculation testing	LOP, incentives, payslip generation
Exam system testing	MCQ creation, student attempts, auto-grading
WhatsApp notification testing	All critical alerts delivered

Client Submission Package

Deliverable	**Content**
Live System Access	Fully functional platform with demo accounts
Demo Video	Complete feature walkthrough
User Guides	Documentation for all 6 roles
API Documentation	Frappe ↔ SmartUp integration specs
Deployment Guide	Server setup and configuration
Testing Reports	Load test results, payment tests, AI accuracy

---

Success Metrics for Feb 28 Delivery

Metric	Target	How to Verify
Concurrent Rooms	5-10 rooms	Load test with 10 rooms active simultaneously
Students Per Room	20 students	Load test with 20 connections to one room
Total Concurrent Students	100 students	Load test with 5 rooms × 20 students each
Video Quality	720p @ 30fps	Visual inspection + technical check
Audio Quality	Clear, no echo/cut	Audio test with multiple participants
Connection Stability	<2% drop rate	Run 60-minute test class
Attendance Accuracy	100% auto-capture	Verify join/leave times match actual
Recording Success (Dual)	100% to MinIO + YouTube	Both storage methods capture all classes
Payment Enforcement	100% blocking	Students with unpaid fees cannot join class
Payment Gateway Success	>95%	Federal Bank transactions succeed
AI Detection Accuracy	>90%	MediaPipe correctly detects drowsiness/distraction
AI Performance	<5% CPU impact	Client-side AI doesn't slow student device

Ghost Mode Invisibility	100%	Ghost admin never appears in participant list
YouTube Stream Stability	<1% drop rate	60-minute stream without interruption
Video Access Approval	100% workflow	Request → Approve → Time-limited link works
Exam Auto-Grading	100% accuracy	MCQ answers graded correctly
WhatsApp Delivery	>95%	Critical notifications reach recipients
Login Success	100%	All 6 roles can log in and access dashboards
Page Load Time	<3 seconds	Dashboard loads within 3s
API Response Time	<500ms	Token generation <500ms

Risks & Mitigations

Risk	Impact	Probability	Mitigation
Server provisioning delay	High	Low	Pre-order servers on Day 1 morning with immediate provisioning
LiveKit configuration issues	High	Medium	Allocate extra time Day 6-7, have LiveKit expert on standby
Load testing	High	Medium	Test with 50 first,

failure (100 students)			optimize, then 100. Have scaling plan ready
Federal Bank gateway integration delays	High	Medium	Start integration Day 1, use test credentials, have backup payment method
MediaPipe performance issues on low-end devices	Medium	Medium	Make AI monitoring optional per batch, optimize detection frequency
YouTube API rate limits	Medium	Low	Request quota increase from Google immediately, implement retry logic
YouTube RTMP stream failures	High	Medium	Always record to MinIO as backup, test YouTube stream Day 7 extensively
WhatsApp API approval delays	Medium	Medium	Use Twilio WhatsApp (faster approval) vs direct WhatsApp Business API
Recording storage running out	Medium	Low	Provision 500GB MinIO initially, YouTube unlimited free backup
Network latency	Medium	Medium	Choose AWS

to GCC			Bahrain or Google Cloud Middle East region
Integration bugs (Frappe-SmartUp-LiveKit)	High	High	Daily integration testing from Day 3 onwards, dedicate 1 person to integration
Browser compatibility (MediaPipe WASM)	Medium	Medium	Test on Chrome, Firefox, Safari. Recommend Chrome for AI features
Aggressive timeline with too many features	High	High	Prioritize ruthlessly: If any feature slips, defer to Phase 2. Core = Live classroom + Payment
Payment gateway PCI compliance	Medium	Low	Use Federal Bank hosted page (they handle PCI compliance)
Ghost Mode detection (students figure it out)	Low	Low	Use LiveKit's native hidden participant feature (protocol-level invisibility)
Exam system question bank data entry	Low	Medium	Provide Excel import template, bulk upload capability
Team burnout	Medium	High	Clear daily goals,

(9-day sprint)			celebrate small wins, avoid overtime beyond Day 8

Client Submission Package

On February 28, deliver:

1. ✅ **Live Demo Access**
   - URL: app.smartup.live (Frappe)
   - URL: class.smartup.live (SmartUp)
   - Demo accounts: 1 Owner, 1 Batch Coordinator, 1 Academic Operator, 2 Teachers, 20 Students, 5 Parents
   - Pre-loaded data: 3 batches, 10 scheduled classes, sample recordings, sample exams
2. ✅ **Demo Video** (10 minutes)
   - **Part 1 (3 min)**: Admin creates batch → Schedules class → Assigns teacher/coordinator
   - **Part 2 (4 min)**: Teacher starts class → Students join after payment → Live classroom with AI monitoring → Ghost Mode demonstration → YouTube live streaming indicator
   - **Part 3 (2 min)**: Student takes MCQ exam → Auto-grading → Results displayed
   - **Part 4 (1 min)**: Student requests video access → Coordinator approves → Time-limited YouTube link received
3. ✅ **Documentation Package**
   - Architecture overview (3-server model, data flow diagrams)
   - User guides for all 6 roles (Owner, Batch Coordinator, Academic Operator, Teacher, Student, Parent)
   - API documentation (Frappe ↔ SmartUp integration, YouTube API, Federal Bank API)
   - Deployment guide (step-by-step server setup, configuration files)
   - Security documentation (payment security, AI privacy, Ghost Mode audit)
4. ✅ **Technical Specifications**
   - Server requirements (CPU, RAM, storage for 100+ students)
   - Network requirements (bandwidth, latency, GCC region optimization)
   - Browser compatibility (Chrome, Firefox, Safari with MediaPipe support)
   - Capacity specs (100 concurrent students verified via load testing)
   - YouTube integration specs (RTMP configuration, API quotas)
   - Payment gateway specs (Federal Bank credentials, multi-currency setup)
5. ✅ **Testing Reports**
   - Load test results (100 concurrent students across 5 rooms)
   - Payment gateway test transactions (all 6 GCC currencies)
   - AI monitoring accuracy report (drowsiness/distraction detection rates)

- YouTube streaming stability report (60-minute stream test)
- Exam system validation (MCQ grading accuracy)
- WhatsApp notification delivery rates

6. ✅ **Phase 2 Roadmap**
   - Feature list with priorities
   - Timeline (March-April 2026)
   - Estimated cost and resource requirements

Conclusion

This **feature-complete Phase 1 delivery** provides a **production-ready live classroom system** by **February 28, 2026** (9 days) with:

✅ **100 concurrent students** across multiple rooms

✅ **6 user roles** with complete workflows (Owner, Batch Coordinator, Academic Operator, Teacher, Student, Parent)

✅ **AI-powered student monitoring** (MediaPipe - drowsiness, distraction, absence detection)

✅ **Ghost Mode for Owner + Batch Coordinator** (quality assurance, silent observation, audit logged)

✅ **Complete HR/Payroll system** (salary structures, LOP, incentives, payslips)

✅ **YouTube live streaming** with approval-based access and dual recording

✅ **Federal Bank payment gateway** with multi-currency GCC and period-wise billing enforcement

✅ **Complete exam system** (MCQ auto-grading, results, report cards, question bank)

✅ **WhatsApp notifications** for critical alerts (class reminders, payments, attendance)

✅ **Dual recording strategy** (MinIO backup + YouTube unlimited storage)

✅ **Real-time attendance** with automatic tracking and parent alerts

✅ **Role-specific dashboards** for all 6 roles with real-time widgets

Thank you